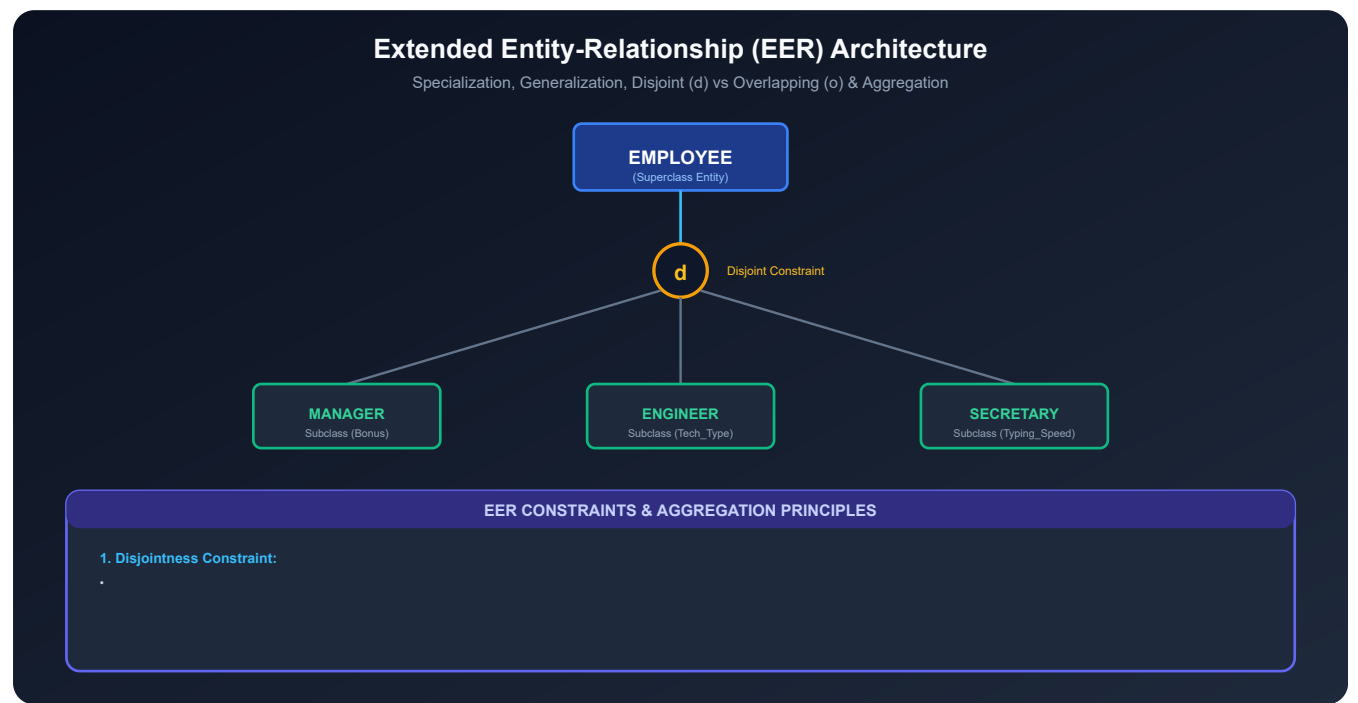


Module 08: Extended ER Features (EER)

1. EER Architecture & Taxonomy



Disjoint (d): Entity can belong to at most ONE subclass (e.g. Can't be Manager AND Secretary simultaneously). • **Overlapping (o):** Entity can belong to MULTIPLE subclasses (e.g. Employee can be Teacher AND Student). 2. Completeness Constraint: • **Total (Double line):** Every superclass entity MUST belong to at least one subclass. • **Partial (Single line):** Some superclass entities may not belong to any subclass. **Aggregation:** Treating a relationship between entity sets as a higher-level composite entity so it can enter a secondary relationship.

2. Specialization vs Generalization

Feature	Specialization	Generalization
Approach	Top-Down Process	Bottom-Up Process
Objective	Identify distinguishing subclass properties	Generalize common superclass attributes
Inheritance	Subclass inherits all superclass attributes	Common attributes extracted into superclass

3. Core EER Constraints

- **Disjoint (d):** Entity can belong to at most 1 subclass.
- **Overlapping (o):** Entity can simultaneously belong to multiple subclasses.
- **Total (Double Line):** Entity must belong to at least 1 subclass.
- **Partial (Single Line):** Entity may not belong to any defined subclass.
- **Aggregation:** Abstracts a relationship into a higher-level composite entity.